

H5: Lie group and Lie algebras: classification and representations	
general idea of Lie group	= groups that depend on a set of continuous parameters that label the group elements > this Lie group is completely characterized by a finite set of generators > these generators form a <i>Lie algebra</i> > Lie group can be obtained from the Lie algebra via the exponential map
5.1 warming up: associativity	
associativity in Lie groups	we want to generalize groups to the continuous case > consider continuous group elements $g(\mathbf{x})$ that depend on a finite number of real continuous variables $\mathbf{x}$ in a certain domain > impose that the group structure is encoded in a continuous differentiable function $\phi(a,b)$ > this function should satisfy the associativity condition: $\phi(a, \phi(b, c)) = \phi(\phi(a, b), c).$ here a, b, c are group elements corresponding to a certain collection of parameters $\vec{x}_a, \vec{x}_b, \vec{x}_c: a \equiv g(\vec{x}_a), b \equiv g(\vec{x}_b), c \equiv g(\vec{x}_c).$
def: SU(2) group	SU(2) is the group of unimodular 2x2 matrices: $SU(2) := \{U \in \mathbb{C}^{2 \times 2} \det U = 1, UU^\dagger = U^\dagger U = \mathbb{1}\}.$ with matrix multiplication the group multiplication
5.2 Baker-Campbell-Hausdorff formula	
exponential map of the Lie group	every element in the Lie group can be written as a matrix exponential of the form: $\exp\left(i \sum_i x_i X_i\right)$ with matrices X_i called the generators of the group the exponential called the exponential map of the generators > the group multiplication is completely determined by: $\exp\left(i \sum_i x_i X_i\right) \exp\left(i \sum_i y_i X_i\right)$ however write $i \sum_i x_i X_i = X, i \sum_i y_i X_i = Y$ then: $\exp(X) \exp(Y).$ in general these don't commute, thus in general: $\exp(X) \exp(Y) \neq \exp(X + Y).$ nevertheless, these can be expressed in terms of commutators of X and Y > this is the <i>Campbell-Baker-Hausdorff</i> formula: $e^X e^Y = e^Z$ with: $Z = X + Y + \frac{1}{2}[X, Y] + \frac{1}{12}[X, [X, Y]] + \frac{1}{12}[Y, [Y, X]] + \dots$
def: Lie algebra $\mathfrak{g}$	= a vector space endowed with a function $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \mapsto \mathfrak{g}$ called the Lie bracket with: 1. <i>bilinear</i> $[\lambda a + b, c + \lambda' d] = \lambda[a, c] + \lambda\lambda'[a, d] + [b, c] + \lambda'[a, d],$ 2. <i>antisymmetric</i> $[a, b] = -[b, a],$ 3. <i>and satisfies the Jacobi identity</i> $[a, [b, c]] + [b, [c, a]] + [c, [a, b]] = 0,$ 0 being the zero vector.
structure factors f	the generators of the Lie group can be thought of as a basis of a Lie algebra > choose such a basis $\mathfrak{g} = \text{span}_{\mathbb{C}} \{X_\alpha\}$ such that we can write down the commutators as follows: $[X_\alpha, X_\beta] = i \sum_{\gamma} f_{\alpha\beta}^{\gamma} X_{\gamma}$ with f the structure factors, that also satisfy the Jacobi identity $\sum_{\mu} f_{\rho\sigma}^{\mu} f_{\mu\tau}^{\phi} + f_{\sigma\tau}^{\mu} f_{\mu\rho}^{\phi} + f_{\tau\rho}^{\mu} f_{\mu\sigma}^{\phi} = 0.$

5.3 representations of Lie algebras	
representation of a Lie algebra	a representation of a Lie algebra is a set of matrices that satisfy the commutation relations: $[X_\alpha, X_\beta] = i \sum_\gamma f_{\alpha\beta}^\gamma X_\gamma.$
adjoint representation	= nontrivial representation of dimension $\dim_{\mathbb{C}} \mathfrak{g}$ defined as: $(X_\alpha)_{\beta\gamma} = f_{\alpha\gamma}^\beta.$ this forms a representation since the structure factors satisfy the Jacobi identity
ex: quantum chromodynamics	quarks and anti-quarks are described by the theory of QCD which has $\mathfrak{su}(3)$ symmetry > quarks exactly transform in the 3D fundamental representation 3 of $\mathfrak{su}(3)$ - anti-quarks transform in the 3D anti-fundamental representation $\bar{3}$ of $\mathfrak{su}(3)$ it can be shown that only bound states of quarks are mesons, 1quark + 1 anti-quark and baryons are 3quarks + 3 anti-quarks > why these combinations? > this is due to $\mathfrak{su}(3)$ symmetry > only these combinations of quarks can transform in the trivial 1 1D representation of $\mathfrak{su}(3)$: $3 \otimes \bar{3} \simeq 1 \oplus 8$ $3 \otimes 3 \otimes 3 \simeq 1 \oplus 8 \oplus 8 \oplus 10.$
5.5 examples of Lie algebras	
def: general linear Lie algebra $\mathfrak{gl}(V)$	= Lie algebra of all complex $n \times n$ matrices with the commutator as Lie bracket $\mathfrak{gl}(V) = \{M \in \mathbb{C}^{n \times n}, n = \dim_{\mathbb{C}}(V)\}$ you can think of $\mathfrak{gl}(V)$ as the endomorphisms of the vector space V > $\mathfrak{gl}(V)$ appears as the Lie algebra associated to the Lie group $GL(n)$
> A family	restrict to the traceless matrices in $\mathfrak{gl}(V)$, then we get the simple Lie algebras: $A_n = \mathfrak{sl}(n+1) = \mathfrak{su}(n+1), n \geq 1$ > these correspond to the Lie algebra of the groups $SU(n)$
> B family	the subalgebras of $\mathfrak{gl}(2n)$ of matrices $M \in \mathfrak{gl}(2n)$ that satisfy $M^T K + K M = 0$ are called the orthogonal subalgebras $\mathfrak{so}(2n)$ or B_n $B_n = \mathfrak{so}(2n+1), n \geq 1$ > correspond to the Lie algebras of the rotation groups in an even number of dimensions Where in the definition K stands for: $\begin{pmatrix} 0_n & \mathbb{1}_n \\ \mathbb{1}_n & 0_n \end{pmatrix}$
> C family	the lie algebra $\mathfrak{sp}(n)$ or C_n correspond to the subalgebra of $\mathfrak{gl}(2n)$ of matrices M for which: $M^T = J M J$ where J is: $\begin{pmatrix} 0_n & \mathbb{1}_n \\ -\mathbb{1}_n & 0_n \end{pmatrix}$ then: $C_n = \mathfrak{sp}(2n), n \geq 1$ >> this series is called the symplectic Lie algebras
> D family	D series are the subalgebras of matrices $M \in \mathfrak{gl}(2n+1)$ that satisfy $M^T K' + K' M = 0$ where K': $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0_n & \mathbb{1}_n \\ 0 & \mathbb{1}_n & 0_n \end{pmatrix}$ thus: $D_n = \mathfrak{so}(2n), n \geq 1$
isomorphism between families	we have: $A_1 \simeq B_1 \simeq C_1 \simeq D_1,$ $D_2 \simeq A_1 \oplus A_1,$ $B_2 \simeq C_2,$ $D_3 \simeq A_3.$

5.6 examples of Lie groups	
def: general linear group $GL(n, \mathbb{F})$	= group of invertible nxn matrices: $GL(n, \mathbb{F}) := \{M \in \mathbb{F}^{n \times n} \det M \neq 0\}.$ the group multiplication is given by matrix multiplication
def: special orthogonal group $SO(n)$	= group of real nxn orthogonal matrices with determinant: $SO(n) := \{M \in \mathbb{R}^{n \times n} MM^T = M^T M = \mathbb{1}, \det M = 1\}.$ > these describe nD rotations
def: orthogonal group $O(n)$	= group of real nxn orthogonal matrices: $O(n) := \{M \in \mathbb{R}^{n \times n} MM^T = M^T M = \mathbb{1}\}.$ > describe nD rotations and reflections
def: unitary group $U(n)$	= group of complex unitary nxn matrices: $U(n) := \{U \in \mathbb{C}^{n \times n} UU^\dagger = U^\dagger U = \mathbb{1}\}$ > due to $\det(AB) = \det(A)\det(B)$ and $\det(A^\dagger) = \overline{\det(A)}$ the determinant of each matrix in $U(n)$ is a phase $e^{i\phi}$
def: special unitary group $SU(n)$	= group of complex unitary nxn matrices with unit determinant: $SU(n) := \{U \in \mathbb{C}^{n \times n} UU^\dagger = U^\dagger U = \mathbb{1}, \det U = 1\} \subset U(n).$
matrix Ω	define the matrix: $\Omega := \begin{pmatrix} 0 & \mathbb{1}_n \\ -\mathbb{1}_n & 0 \end{pmatrix}$
def: (complex) symplectic groups $Sp(2n, \mathbb{C})$	these are the matrices defined as: $Sp(2n, \mathbb{C}) := \{M \in \mathbb{C}^{2n \times 2n} M^\top \Omega M = \Omega\}$
def: compact symplectic groups $USp(2n)$	these are defined as: $USp(2n) = Sp(2n, \mathbb{C}) \cap U(2n).$ which correspond to the Lie algebra $\mathfrak{sp}(2n)$